

# STA 540L Case Study 1: Reproduction

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## Filtering out extra rows

```
df <- read.csv("CTN_FINAL.csv")
df[1:6, 1:6]
```

```
##      Q68_FU20 Q31_FU22 Q59_FU23 Q61_FU23 Q78_FU23 Q79_FU23
## 1          NA        NA        NA        NA        NA        NA
## 2          NA        NA        NA        NA        NA        NA
## 3          9999        NA        NA        NA        NA        NA
## 4          NA        NA        NA        NA        NA        NA
## 5          9999        NA        NA        NA        NA        NA
## 6          NA        NA        NA        NA        NA        NA
```

```
dim(df)
```

```
## [1] 271 478
```

The PO\_FLAG variable marks which rows are ultimately used for analysis. However there is 1 additional row unaccounted for by this variable, which I removed by process of elimination based on matching numbers with the original Table 1.

```
# df2 <- subset(df, (Q6_1 != 2) & (LAST_HIV_TEST > 90) & (Q6_2 != 2))
df2 <- df[df$PO_FLAG == "Include", ]
dim(df2)
```

```
## [1] 255 478
```

```
# Decided to manually remove this row based on matching it to the Table 1's stats (not
  ↳ sure why it was thrown out though but whatever):

# df2[df2$Q5_1 == 1 & df2$Q6_2 == 3 & df2$Q11_3 == 1 & df2$Q11_4 == 1 & df2$Q11_5 == 2 &
  ↳ df2$Q11_7 == 3, ]
# which(df2$Q5_1 == 1 & df2$Q6_2 == 3 & df2$Q11_3 == 1 & df2$Q11_4 == 1 & df2$Q11_5 == 2
  ↳ & df2$Q11_7 == 3)

# df2[109, ]$Q3_1
df2 <- df2[-109, ]
dim(df2)
```

```
## [1] 254 478
```

```
# this person's response is bugged, they said they have other reasons for not taking HIV  
↪ test but have NA for LAST_HIV_TEST but also have 1 (Yes) for "Did you take HIV test".  
↪ However, they are supposed to be included in the final analysis.  
# They should be counted as having taken the HIV test and excluded from the "Other" count  
↪ of reasons not to take the HIV test.  
# which(df2$Q11_7 == 8 & is.na(df2$LAST_HIV_TEST))  
# df2[132, ]$Q11_5  
# df2[132, ]$LAST_HIV_TEST_MONTHS
```

## Table 1

```
title <- paste0("Summary of National Institute on Drug Abuse Clinical Trials Network  
↪ Social Media Pre-exposure Prophylaxis Study, 2020, population sociodemographic and  
↪ behavioral characteristics (N=", dim(df2)[1], ").")
```

```
df_race <- factor(df2$Q5_3,  
  levels = c("25", "26",  
             "24", "27", "23", "28",  
             "24,23", "26,24", "25,24,27,23", "24,28", "25,24,23", "25,24"  
             ),  
  labels = c("American Indian or Alaska Native", "Asian",  
             "Black or African American",  
             "Native Hawaiian or Pacific Islander",  
             "White",  
             "Other",  
             "Multiracial", "Multiracial", "Multiracial",  
             "Multiracial", "Multiracial", "Multiracial"))
```

```
age_tbl <- data.frame(  
  Characteristic = c(  
    "Age in Years, median (IQR)",  
    "Ethnicity, n (%)",  
    "Hispanic/Latinx"  
  ),  
  Value = c(  
    paste0(  
      quantile(df2$Q3_1, 0.5), " ("  
      quantile(df2$Q3_1, 0.25), "-"  
      quantile(df2$Q3_1, 0.75), ")"  
    ),  
    "",  
    paste0(  
      sum(df2$Q5_1 == 1), " ("  
      round(100 * sum(df2$Q5_1 == 1) / length(df2$Q5_1), 1), ")"  
    )  
  )  
)
```

```
## NOTE: The race section excluded 4 people out of the 254 participants (they had NA for  
↪ race), and percentages are out of 250
```

```

race_tbl <- data.frame(
  Characteristic = c(
    "Race, n (%)",
    "American Indian or Alaskan Native",
    "Black or African American",
    "White",
    "Other",
    "Multiracial"
  ),
  Value = c(
    "",
    paste0(sum(df_race == "American Indian or Alaska Native", na.rm=TRUE), " (",
      round(100 * sum(df_race == "American Indian or Alaska Native", na.rm=TRUE) /
        ↪ sum(!is.na(df_race)), 1), ")"),
    paste0(sum(df_race == "Black or African American", na.rm=TRUE), " (",
      round(100 * sum(df_race == "Black or African American", na.rm=TRUE) /
        ↪ sum(!is.na(df_race)), 1), ")"),
    paste0(sum(df_race == "White", na.rm=TRUE), " (",
      round(100 * sum(df_race == "White", na.rm=TRUE) / sum(!is.na(df_race)), 1),
        ↪ ")"),
    paste0(sum(df_race == "Other", na.rm=TRUE), " (",
      round(100 * sum(df_race == "Other", na.rm=TRUE) / sum(!is.na(df_race)), 1),
        ↪ ")"),
    paste0(sum(df_race == "Multiracial", na.rm=TRUE), " (",
      round(100 * sum(df_race == "Multiracial", na.rm=TRUE) / sum(!is.na(df_race)),
        ↪ 1), ")")
  )
)

prep_table <- data.frame(
  Characteristic = c(
    "History of PrEP uptake, n (%)",
    "Never taken PrEP",
    "Have taken PrEP, but not within the past 6 months"
  ),
  Value = c(
    "",
    paste0(sum(df2$Q6_2 == 3), " (",
      round(100 * sum(df2$Q6_2 == 3) / dim(df2)[1], 1), ")"),
    paste0(sum(df2$Q6_2 == 1), " (",
      round(100 * sum(df2$Q6_2 == 1) / dim(df2)[1], 1), ")")
  )
)

mpart_table <- data.frame(
  Characteristic = "Number of male sex partners in the past 90 days, median (IQR)",
  Value = paste0(
    quantile(df2$Q11_2, 0.5), " (",
    quantile(df2$Q11_2, 0.25), "-",
    quantile(df2$Q11_2, 0.75), ")"
  )
)

condom_use <- factor(df2$Q11_3, levels = c(1, 2, 3, 4, 5), labels = c("Never",
  ↪ "Sometimes", "About half the time", "Most of the time", "Always"))

```

```

condom_table <- data.frame(
  Characteristic = c(
    "Condom use, n (%)",
    "Never",
    "Sometimes",
    "About half the time",
    "Most of the time",
    "Always",
    "Condomless receptive anal sex in the past 90 days, n (%)"
  ),
  Value = c(
    "",
    paste0(sum(condom_use == "Never"), " (",
      round(100 * sum(condom_use == "Never") / length(condom_use), 1), ")"),
    paste0(sum(condom_use == "Sometimes"), " (",
      round(100 * sum(condom_use == "Sometimes") / length(condom_use), 1), ")"),
    paste0(sum(condom_use == "About half the time"), " (",
      round(100 * sum(condom_use == "About half the time") / length(condom_use), 1),
      ↪ ")"),
    paste0(sum(condom_use == "Most of the time"), " (",
      round(100 * sum(condom_use == "Most of the time") / length(condom_use), 1),
      ↪ ")"),
    paste0(sum(condom_use == "Always"), " (",
      round(100 * sum(condom_use == "Always") / length(condom_use), 1), ")"),
    paste0(sum(df2$Q11_4 == 1), " (",
      round(100 * sum(df2$Q11_4 == 1) / dim(df2)[1], 1), ")")
  )
)

# df2$LAST_HIV_TEST_MONTHS[df2$LAST_HIV_TEST_MONTHS == 9999] <- NA # 9999 should mean
↪ they didn't take the HIV test so set as NA to ignore
tested_for_HIV <- df2[df2$Q11_5 == 1, ] # 191 rows
not_tested <- df2[df2$Q11_5 == 2, ] # 63 rows

HIV_tested_table <- data.frame(
  Characteristic = c(
    "Ever tested for HIV during lifetime, n (%)",
    "If tested for HIV, median (IQR)",
    "Months since last HIV test",
    "If not tested for HIV, n (%)"
  ),
  Value = c(
    paste0(sum(df2$Q11_5 == 1), " (",
      round(100 * sum(df2$Q11_5 == 1) / dim(df2)[1], 1), ")"),
    "",
    paste0(
      round(quantile(tested_for_HIV$LAST_HIV_TEST_MONTHS, 0.5, na.rm = TRUE), 1),
      " (",
      round(quantile(tested_for_HIV$LAST_HIV_TEST_MONTHS, 0.25, na.rm = TRUE), 1),
      "-",
      round(quantile(tested_for_HIV$LAST_HIV_TEST_MONTHS, 0.75, na.rm = TRUE), 1),
      ")"
    ),
    ""
  )
)

```

```

    paste0(sum(df2$Q11_5 == 2), " (",
            round(100 * (sum(df2$Q11_5 == 2)) / dim(df2)[1], 1), "%)"
  )
)

reason_factor <- factor(not_tested$Q11_7, levels = c(1, 2, 3, 4, 5, 6, 7, 8), labels =
  ↪ c("Unlikely to be exposed to HIV", "Afraid of testing HIV-positive", "Did not want to
  ↪ think about HIV/HIV-positive", "Worried about names being reported if positive",
  ↪ "Dislike for needles", "Unable to trust that the results will be confidential",
  ↪ "Unaware of where to get tested", "Other reasons"))

reason_table <- data.frame(
  Characteristic = c(
    "Main reasons cited by the 63 participants for not getting tested, n (%)",
    "Unlikely to be exposed to HIV",
    "Afraid of testing HIV-positive",
    "Did not want to think about HIV/HIV-positive",
    "Worried about names being reported if positive",
    "Dislike for needles",
    "Unable to trust that the results will be confidential",
    "Unaware of where to get tested",
    "Other reasons"
  ),
  Value = c(
    "",
    paste0(sum(reason_factor == "Unlikely to be exposed to HIV"), " (",
            round(100 * sum(reason_factor == "Unlikely to be exposed to HIV") /
              ↪ sum(df2$Q11_5 == 2), 1), "%)"
    ),
    paste0(sum(reason_factor == "Afraid of testing HIV-positive"), " (",
            round(100 * sum(reason_factor == "Afraid of testing HIV-positive") /
              ↪ sum(df2$Q11_5 == 2), 1), "%)"
    ),
    paste0(sum(reason_factor == "Did not want to think about HIV/HIV-positive"), " (",
            round(100 * sum(reason_factor == "Did not want to think about
              ↪ HIV/HIV-positive") / sum(df2$Q11_5 == 2), 1), "%)"
    ),
    paste0(sum(reason_factor == "Worried about names being reported if positive"), " (",
            round(100 * sum(reason_factor == "Worried about names being reported if
              ↪ positive") / sum(df2$Q11_5 == 2), 1), "%)"
    ),
    paste0(sum(reason_factor == "Dislike for needles"), " (",
            round(100 * sum(reason_factor == "Dislike for needles") / sum(df2$Q11_5 == 2),
              ↪ 1), "%)"
    ),
    paste0(sum(reason_factor == "Unable to trust that the results will be confidential"),
            ↪ " (",
            round(100 * sum(reason_factor == "Unable to trust that the results will be
              ↪ confidential") / sum(df2$Q11_5 == 2), 1), "%)"
    ),
    paste0(sum(reason_factor == "Unaware of where to get tested"), " (",
            round(100 * sum(reason_factor == "Unaware of where to get tested") /
              ↪ sum(df2$Q11_5 == 2), 1), "%)"
    ),
    paste0(sum(reason_factor == "Other reasons"), " (",
            round(100 * sum(reason_factor == "Other reasons") / sum(df2$Q11_5 == 2), 1),
              ↪ "%)"
    )
  )
)

```

```

table1_df <- rbind(
  age_tbl,
  race_tbl,
  prep_table,
  mpart_table,
  condom_table,
  HIV_tested_table,
  reason_table
)

table1_df %>%
  kable(caption = title) %>%
  add_indent(positions = c(3)) %>%
  add_indent(positions = 5:9) %>%
  add_indent(positions = 11:12) %>%
  add_indent(positions = 15:19) %>%
  add_indent(positions = 23) %>%
  add_indent(positions = 26:33)

```

Table 1: Summary of National Institute on Drug Abuse Clinical Trials Network Social Media Pre-exposure Prophylaxis Study, 2020, population sociodemographic and behavioral characteristics (N=254).

| Characteristic                                                          | Value           |
|-------------------------------------------------------------------------|-----------------|
| Age in Years, median (IQR)                                              | 25 (21-27)      |
| Ethnicity, n (%)                                                        |                 |
| Hispanic/Latinx                                                         | 66 (26)         |
| Race, n (%)                                                             |                 |
| American Indian or Alaskan Native                                       | 1 (0.4)         |
| Black or African American                                               | 196 (78.4)      |
| White                                                                   | 28 (11.2)       |
| Other                                                                   | 14 (5.6)        |
| Multiracial                                                             | 11 (4.4)        |
| History of PrEP uptake, n (%)                                           |                 |
| Never taken PrEP                                                        | 232 (91.3)      |
| Have taken PrEP, but not within the past 6 months                       | 22 (8.7)        |
| Number of male sex partners in the past 90 days, median (IQR)           | 4 (3-6)         |
| Condom use, n (%)                                                       |                 |
| Never                                                                   | 36 (14.2)       |
| Sometimes                                                               | 108 (42.5)      |
| About half the time                                                     | 37 (14.6)       |
| Most of the time                                                        | 68 (26.8)       |
| Always                                                                  | 5 (2)           |
| Condomless receptive anal sex in the past 90 days, n (%)                | 210 (82.7)      |
| Ever tested for HIV during lifetime, n (%)                              | 191 (75.2)      |
| If tested for HIV, median (IQR)                                         |                 |
| Months since last HIV test                                              | 10.9 (6.1-21.1) |
| If not tested for HIV, n (%)                                            | 63 (24.8)       |
| Main reasons cited by the 63 participants for not getting tested, n (%) |                 |

|                                                       |           |
|-------------------------------------------------------|-----------|
| Unlikely to be exposed to HIV                         | 8 (12.7)  |
| Afraid of testing HIV-positive                        | 26 (41.3) |
| Did not want to think about HIV/HIV-positive          | 8 (12.7)  |
| Worried about names being reported if positive        | 3 (4.8)   |
| Dislike for needles                                   | 5 (7.9)   |
| Unable to trust that the results will be confidential | 3 (4.8)   |
| Unaware of where to get tested                        | 7 (11.1)  |
| Other reasons                                         | 3 (4.8)   |

---

## Primary analysis

```
library(emmeans)
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```
df2$TIME_WAVE_ACTIVE <- ifelse(df2$SITE == "Facebook" | df2$SITE == "Google" | df2$SITE
  ↳ == "Grindr", 70, 38)
df2$WAVE_ADJ <- ifelse(df2$WAVE == 4 | df2$WAVE == 1, 1, 2)
# compile number of test kits ordered per site
# sum(df2[df2$SITE == "Facebook", ]$ORA_REDEEMED == "Yes") # number of ppl who redeemed
  ↳ test kit from facebook
df2$SITE_FACTOR <- factor(df2$SITE, levels = c("Facebook", "Grindr", "Google",
  ↳ "Instagram", "Jack'd", "Bing"))
df2$SITE_TYPE <- factor(df2$SITE_TYPE, levels = c("Dating Apps", "Info Sites", "Social
  ↳ Media"))
```

```
# df2$SITE_TYPE # platform name
# df2$SITE # site name
# df2$WAVE # wave number: wave #4 has same sites as wave #1
# sum(df2$ORA_REDEEMED == "Yes") # number of ppl who redeemed test kit = 177
# df2$ORA_WITHIN60 # days since survey start date that ppl redeemed test kit, NA if not
  ↳ redeemed
# sum(df2[df2$SITE == "Facebook", ]$ORA_REDEEMED == "Yes") # number of ppl who redeemed
  ↳ test kit from facebook
```

## Fitting Poisson Model

```
df2[df2$SITE_TYPE == "Dating Apps" & df2$SITE=="Instagram", ]$SITE_TYPE <- "Social Media"
  ↳ ## 4 of the instagram columns are labeled as being dating apps but they're not
```

```
df_model_raw <- df2 %>%
  group_by(SITE_TYPE, WAVE_ADJ) %>%
  summarise(
    outcome = sum(ORA_REDEEMED == "Yes"),
```

```

    TIME_WAVE_ACTIVE = first(TIME_WAVE_ACTIVE),
    .groups = "drop"
  )
df_model <- rbind(df_model_raw, data.frame(
  "SITE_TYPE" = "Info Sites", "WAVE_ADJ" = 2,
  "outcome" = 0, "TIME_WAVE_ACTIVE" = 38
))
) # add bing

df_model$WAVE_ADJ <- factor(df_model$WAVE_ADJ, levels=c(1, 2))
df_model$SITE_NAME <- c("Grindr", "Jack'd", "Google",
  "Facebook", "Instagram", "Bing")

primary_fit <- glm(
  outcome ~ SITE_TYPE * WAVE_ADJ + offset(log(TIME_WAVE_ACTIVE)),
  family = poisson,
  data = df_model
)

summary(primary_fit)

##
## Call:
## glm(formula = outcome ~ SITE_TYPE * WAVE_ADJ + offset(log(TIME_WAVE_ACTIVE)),
##      family = poisson, data = df_model)
##
## Coefficients:
##                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)          -2.0513     0.3333  -6.154 7.56e-10 ***
## SITE_TYPEInfo Sites      0.6360     0.4122   1.543   0.123
## SITE_TYPESocial Media    0.3677     0.4336   0.848   0.396
## WAVE_ADJ2              3.2420     0.3451   9.394 < 2e-16 ***
## SITE_TYPEInfo Sites:WAVE_ADJ2 -27.7669 42247.1657 -0.001   0.999
## SITE_TYPESocial Media:WAVE_ADJ2 -2.6311     0.5225 -5.036 4.75e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 3.5500e+02  on 5  degrees of freedom
## Residual deviance: 4.1223e-10  on 0  degrees of freedom
## AIC: 36.233
##
## Number of Fisher Scoring iterations: 20

```

## Estimating cell-specific rates

```

rate_ests_emmeans <- emmeans(
  primary_fit,
  ~ SITE_TYPE * WAVE_ADJ,

```

```

offset = 0,
type = "response"
)

## NOTE: A nesting structure was detected in the fitted model:
##      WAVE_ADJ %in% TIME_WAVE_ACTIVE

rate_ests <- data.frame(rate_ests_emmeans)
site_names <- data.frame(
  "SITE_TYPE" = c("Dating Apps", "Dating Apps",
                  "Social Media", "Social Media",
                  "Info Sites", "Info Sites"),
  "WAVE_ADJ" = factor(c(1,2,1,2,1,2), levels=c(1,2)),
  "SITE_NAME" = c("Grindr", "Jack'd", "Facebook",
                  "Instagram", "Google", "Bing")
)
rate_ests <- rate_ests %>%
  left_join(site_names, by = c("SITE_TYPE", "WAVE_ADJ")) %>%
  mutate(
    SITE_TYPE = factor(
      SITE_TYPE,
      levels = c("Social Media", "Dating Apps", "Info Sites")
    ),
    WAVE_ADJ = factor(WAVE_ADJ, levels = c(1, 2))
  ) %>%
  arrange(SITE_TYPE, WAVE_ADJ)

kable(rate_ests[, c(9,1,2,3,4, 5)] %>%
  mutate(
    rate = round(rate, 2),
    SE = round(SE, 2)
  ),
  col.names = c("Site Name", "Wave", "Number of days per wave", "Platform Type",
    ↪ "Rate", "95% SE"))

```

| Site Name | Wave | Number of days per wave | Platform Type | Rate | 95% SE |
|-----------|------|-------------------------|---------------|------|--------|
| Facebook  | 1    | 70                      | Social Media  | 0.19 | 0.05   |
| Instagram | 2    | 38                      | Social Media  | 0.34 | 0.09   |
| Grindr    | 1    | 70                      | Dating Apps   | 0.13 | 0.04   |
| Jack'd    | 2    | 38                      | Dating Apps   | 3.29 | 0.29   |
| Google    | 1    | 70                      | Info Sites    | 0.24 | 0.06   |
| Bing      | 2    | 38                      | Info Sites    | 0.00 | 0.00   |

## Contrasts

```

df_model2 <- rbind(df_model_raw, data.frame(
  "SITE_TYPE" = "Info Sites", "WAVE_ADJ" = 2,
  "outcome" = 0.5, "TIME_WAVE_ACTIVE" = 38
))

```

```

) # add bing with small pseudo count
df_model2$WAVE_ADJ <- factor(df_model2$WAVE_ADJ, levels=c(1, 2))
df_model2$SITE_NAME <- c("Grindr", "Jack'd", "Google",
                         "Facebook", "Instagram", "Bing")
primary_fit2 <- glm(
  outcome ~ SITE_TYPE * WAVE_ADJ + offset(log(TIME_WAVE_ACTIVE)),
  family = poisson,
  data = df_model2
)

rate_ests_emm_pseudo <- emmeans(
  primary_fit2,
  ~ SITE_TYPE * WAVE_ADJ,
  offset = 0,
  type = "response"
)

```

```

## NOTE: A nesting structure was detected in the fitted model:
##      WAVE_ADJ %in% TIME_WAVE_ACTIVE

```

```

contrasts_pvals <- data.frame(contrast(rate_ests_emm_pseudo,
  method="pairwise",
  by="WAVE_ADJ",
  type="response",
  adjust="hochberg"))

```

```

## Note: Grouping factor(s) for 'WAVE_ADJ' have been added to the 'by' list.

```

```

contrasts_pvals$platform_names <- c(
  "Jack'd vs Bing",
  "Jack'd vs Instagram",
  "Bing vs Instagram",
  "Grindr vs Google",
  "Grindr vs Facebook",
  "Google vs Facebook"
)

kable(contrasts_pvals[, c(1, 10, 9)] %>%
  mutate(
    p.value = round(p.value, 4),
  ) %>%
  arrange(WAVE_ADJ),
  col.names = c("Wave", "Contrast", "P-value"))

```

| Wave | Contrast            | P-value |
|------|---------------------|---------|
| 1    | Grindr vs Google    | 0.3686  |
| 1    | Grindr vs Facebook  | 0.4665  |
| 1    | Google vs Facebook  | 0.4665  |
| 2    | Jack'd vs Bing      | 0.0002  |
| 2    | Jack'd vs Instagram | 0.0000  |

| Wave | Contrast          | P-value |
|------|-------------------|---------|
| 2    | Bing vs Instagram | 0.0238  |

## Secondary analysis

### a) Tobacco, alcohol, prescription medications, and other substance use

```
# which(df2$ORA_REDEEMED != "Yes" & (df2$ORA_REDEEMED != "No"))
df2[118, ]$ORA_REDEEMED <- "No" # there was an "Over 60 Days" but in the analysis should
↪ count as No

# None: 0 pts, Problem use: 1 pts, High risk use: > 1 pts

# Alcohol
df2$alc_score <-
  rowSums(df2[, c("Q13_1", "Q13_2", "Q13_3", "Q13_4")] == 1, na.rm=TRUE)
df2$alc_score_str <- ifelse(
  df2$alc_score == 0, "None",
  ifelse(
    df2$alc_score == 1, "Problem Use",
    "High Risk Substance Use"
  )
)
# Cannabis
df2$can_score <-
  rowSums(df2[, c("Q13_5", "Q13_6", "Q13_7")] == 1, na.rm=TRUE)
df2$can_score_str <- ifelse(
  df2$can_score == 0, "None",
  ifelse(
    df2$can_score == 1, "Problem Use",
    "High Risk Substance Use"
  )
)
# Stimulants - all below collapse problem use and high risk into 1 category
df2$stim_score <-
  rowSums(df2[, c("Q13_8", "Q13_9", "Q13_10")] == 1, na.rm=TRUE)
df2$stim_score_str <- ifelse(df2$stim_score < 1, "None", "Problem Use/High Risk Substance
↪ Use")

# opioid
df2$opioid_score <-
  rowSums(
    df2[, c(
      "Q13_11", "Q13_12", "Q13_13",
      "Q13_14", "Q13_15", "Q13_16"
    )] == 1,
    na.rm = TRUE
  )
df2$opioid_score_str <- ifelse(df2$opioid_score < 1, "None", "Problem Use/High Risk
↪ Substance Use")
```

```

# sedative
df2$sedative_score <-
  rowSums(df2[, c("Q13_17", "Q13_18", "Q13_19")] == 1, na.rm=TRUE)
df2$sedative_score_str <- ifelse(df2$sedative_score < 1, "None", "Problem Use/High Risk
  ↳ Substance Use")

# prescribed stimulant
df2$rx_score <-
  rowSums(df2[, c("Q13_20", "Q13_21", "Q13_22")] == 1)
df2$rx_score_str <- ifelse(df2$rx_score < 1, "None", "Problem Use/High Risk Substance
  ↳ Use")

# check
# sum(df2$sedative_score == 0 & df2$ORA_REDEEMED == "Yes")
# sum(df2$sedative_score == 1 & df2$ORA_REDEEMED == "Yes")
# sum(df2$sedative_score > 1 & df2$ORA_REDEEMED == "Yes")
# sum(df2$sedative_score == 0 & df2$ORA_REDEEMED == "No")
# sum(df2$sedative_score == 1 & df2$ORA_REDEEMED == "No")
# sum(df2$sedative_score > 1 & df2$ORA_REDEEMED == "No")

## p value tests

# Alcohol
alcohol_pval <- fisher.test(data.frame(
  rbind(table(df2[df2$ORA_REDEEMED == "Yes", ]$alc_score_str),
    table(df2[df2$ORA_REDEEMED == "No", ]$alc_score_str)),
  row.names = c("Ordered test kit", "Did not order test kit")
))
)p.value

# Cannabis
can_pval <- fisher.test(data.frame(
  rbind(table(df2[df2$ORA_REDEEMED == "Yes", ]$can_score_str),
    table(df2[df2$ORA_REDEEMED == "No", ]$can_score_str)),
  row.names = c("Ordered test kit", "Did not order test kit")
))
)p.value

# Stimulants
stim_pval <- fisher.test(data.frame(
  rbind(table(df2[df2$ORA_REDEEMED == "Yes", ]$stim_score_str),
    table(df2[df2$ORA_REDEEMED == "No", ]$stim_score_str)),
  row.names = c("Ordered test kit", "Did not order test kit")
))
)p.value

# Opioids
opioid_pval <- fisher.test(data.frame(
  rbind(table(df2[df2$ORA_REDEEMED == "Yes", ]$opioid_score_str),
    table(df2[df2$ORA_REDEEMED == "No", ]$opioid_score_str)),
  row.names = c("Ordered test kit", "Did not order test kit")
))
)p.value

```

```

# Sedative
sed_pval <- fisher.test(data.frame(
  rbind(table(df2[df2$ORA_REDEEMED == "Yes", ]$sedative_score_str),
    table(df2[df2$ORA_REDEEMED == "No", ]$sedative_score_str)),
  row.names = c("Ordered test kit", "Did not order test kit")
)
)$p.value

# Prescribed stim
rx_stim_pval <- fisher.test(data.frame(
  rbind(table(df2[df2$ORA_REDEEMED == "Yes", ]$rx_score_str),
    table(df2[df2$ORA_REDEEMED == "No", ]$rx_score_str)),
  row.names = c("Ordered test kit", "Did not order test kit")
)
)$p.value

# list with names of substances and associated p values from before
substances <- list(
  Alcohol = list(var = "alc_score_str", pval = alcohol_pval),
  Cannabis = list(var = "can_score_str", pval = can_pval),
  Stimulants = list(var = "stim_score_str", pval = stim_pval),
  Opioid = list(var = "opioid_score_str", pval = opioid_pval),
  Sedative = list(var = "sedative_score_str", pval = sed_pval),
  `Prescribed Stimulant` = list(var = "rx_score_str", pval = rx_stim_pval)
)

# initialize results dataframe with column names
results_df <- data.frame(
  Substance = character(),
  Category = character(),
  Count_Yes = integer(),
  Percent_Yes = numeric(),
  Count_No = integer(),
  Percent_No = numeric(),
  P_value = numeric(),
  stringsAsFactors = FALSE
)

for (sub in names(substances)) { # repeat per substance
  var <- substances[[sub]]$var
  p_val <- substances[[sub]]$pval

  if (sub %in% c("Stimulants", "Opioid", "Sedative", "Prescribed Stimulant")) {
    df2[[var]] <- ifelse(df2[[var]] == "None", "None", "Problem Use/High Risk Substance
    ↪ Use") # collapsed problem use and high risk categories
    category_order <- c("None", "Problem Use/High Risk Substance Use")
  } else { # separated problem use and high risk categories
    category_order <- c("None", "Problem Use", "High Risk Substance Use")
  }

  tab <- table(df2[[var]], df2$ORA_REDEEMED)

  first_row <- TRUE

```

```

for (cat in category_order) {
  if (!(cat %in% rownames(tab))) {
    count_yes <- 0
    count_no <- 0
  } else {
    count_yes <- tab[cat, "Yes"]
    count_no <- tab[cat, "No"]
  }

  percent_yes <- round(100 * count_yes / sum(tab[, "Yes"]), 1)
  percent_no <- round(100 * count_no / sum(tab[, "No"]), 1)

  pval_to_use <- if (first_row) round(p_val, 3) else ""
  sub_to_use <- if (first_row) sub else ""

  results_df <- rbind(
    results_df,
    data.frame(
      Substance = sub_to_use,
      Category = cat,
      Count_Yes = count_yes,
      Percent_Yes = percent_yes,
      Count_No = count_no,
      Percent_No = percent_no,
      P_value = pval_to_use,
      stringsAsFactors = FALSE
    )
  )
  first_row <- FALSE
}
}

kable(results_df,
  col.names =
    c("Substance", "Category", "Ordered test kit",
      "Percent", "Did not order test kit", "Percent",
      "P-value"))

```

| Substance  | Category                               | Ordered test<br>kit | Percent | Did not order<br>test kit | Percent | P-<br>value |
|------------|----------------------------------------|---------------------|---------|---------------------------|---------|-------------|
| Alcohol    | None                                   | 52                  | 29.4    | 16                        | 20.8    | 0.35        |
|            | Problem Use                            | 46                  | 26.0    | 21                        | 27.3    |             |
|            | High Risk Substance Use                | 79                  | 44.6    | 40                        | 51.9    |             |
| Cannabis   | None                                   | 89                  | 50.3    | 32                        | 41.6    | 0.096       |
|            | Problem Use                            | 37                  | 20.9    | 12                        | 15.6    |             |
|            | High Risk Substance Use                | 51                  | 28.8    | 33                        | 42.9    |             |
| Stimulants | None                                   | 166                 | 93.8    | 69                        | 89.6    | 0.299       |
|            | Problem Use/High Risk<br>Substance Use | 11                  | 6.2     | 8                         | 10.4    |             |
|            |                                        |                     |         |                           |         |             |
| Opioid     | None                                   | 173                 | 97.7    | 75                        | 97.4    | 1           |
|            | Problem Use/High Risk<br>Substance Use | 4                   | 2.3     | 2                         | 2.6     |             |
|            |                                        |                     |         |                           |         |             |
| Sedative   | None                                   | 168                 | 94.9    | 72                        | 93.5    | 0.766       |

| Substance               | Category                               | Ordered test<br>kit | Percent | Did not order<br>test kit | Percent | P-<br>value |
|-------------------------|----------------------------------------|---------------------|---------|---------------------------|---------|-------------|
| Prescribed<br>Stimulant | Problem Use/High Risk<br>Substance Use | 9                   | 5.1     | 5                         | 6.5     | 0.702       |
|                         | None                                   | 172                 | 97.2    | 74                        | 96.1    |             |
|                         | Problem Use/High Risk<br>Substance Use | 5                   | 2.8     | 3                         | 3.9     |             |

## b) Stage of Health Behavior Change

```
df2 <- df2 %>%
  mutate(
    stage_behav_change = case_when(
      Q15_1 == 1 ~ "Precontemplation",
      Q15_1 == 2 ~ "Contemplation",
      Q15_1 == 3 ~ "Determination",
      Q15_1 == 4 ~ "Action",
      Q15_1 == 5 ~ "Maintenance"
    )
  )

stage_behav_pval <- fisher.test(data.frame(
  rbind(table(df2[df2$ORA_REDEEMED == "Yes", ]$stage_behav_change),
        table(df2[df2$ORA_REDEEMED == "No", ]$stage_behav_change)),
  row.names = c("Ordered test kit", "Did not order test kit")
))
)p.value
```

```
stage_levels <- c(
  "Precontemplation",
  "Contemplation",
  "Determination",
  "Action",
  "Maintenance"
)

stage_labels <- c(
  "I do not see any need to regularly test for HIV",
  "I think I should get tested for HIV regularly, but I am not sure",
  "I am ready to start getting regularly tested for HIV",
  "I am trying to get tested regularly for HIV",
  "I have been getting tested for HIV regularly over the past few years"
)

results_stage <- data.frame(
  Stage = character(),
  Description = character(),
  Count_Yes = integer(),
  Percent_Yes = numeric(),
  Count_No = integer(),
```

```

Percent_No = numeric(),
P_value = character(),
stringsAsFactors = FALSE
)

tab <- table(df2$stage_behav_change, df2$ORA_REDEEMED)

first_row <- TRUE
for (i in seq_along(stage_levels)) {
  lvl <- stage_levels[i]

  count_yes <- ifelse(lvl %in% rownames(tab), tab[lvl, "Yes"], 0)
  count_no <- ifelse(lvl %in% rownames(tab), tab[lvl, "No"], 0)

  percent_yes <- round(100 * count_yes / sum(tab[, "Yes"]), 1)
  percent_no <- round(100 * count_no / sum(tab[, "No"]), 1)

  results_stage <- rbind(
    results_stage,
    data.frame(
      Stage = stage_levels[i],
      Description = stage_labels[i],
      Count_Yes = count_yes,
      Percent_Yes = percent_yes,
      Count_No = count_no,
      Percent_No = percent_no,
      P_value = if (first_row) round(stage_behav_pval, 3) else "",
      stringsAsFactors = FALSE
    )
  )
  first_row <- FALSE
}

kable(
  results_stage,
  col.names = c(
    "Attitude", "Description",
    "Ordered test kit", "Percent",
    "Did not order test kit", "Percent",
    "P-value"
  )
)

```

| Attitude         | Description                                                      | Ordered<br>test kit | Percent | Did not order<br>test kit | Percent | P-<br>value |
|------------------|------------------------------------------------------------------|---------------------|---------|---------------------------|---------|-------------|
| Precontemplation | I do not see any need to regularly test for HIV                  | 7                   | 4.0     | 2                         | 2.6     | 0.251       |
| Contemplation    | I think I should get tested for HIV regularly, but I am not sure | 31                  | 17.5    | 8                         | 10.4    |             |
| Determination    | I am ready to start getting regularly tested for HIV             | 87                  | 49.2    | 49                        | 63.6    |             |
| Action           | I am trying to get tested regularly for HIV                      | 40                  | 22.6    | 12                        | 15.6    |             |

| Attitude    | Description                                                             | Ordered<br>test kit | Percent | Did not order<br>test kit | P-<br>Percentvalue |
|-------------|-------------------------------------------------------------------------|---------------------|---------|---------------------------|--------------------|
| Maintenance | I have been getting tested for HIV<br>regularly over the past few years | 12                  | 6.8     | 6                         | 7.8                |

### c) Attitudes toward human immunodeficiency virus (HIV) testing

```
df2$test_attitudes1 <- ifelse(df2$Q15_3 == 1, "Agree", "Disagree")
df2$test_attitudes2 <- ifelse(df2$Q15_4 == 1, "Agree", "Disagree")
df2$test_attitudes3 <- ifelse(df2$Q15_5 == 1, "Agree", "Disagree")
df2$test_attitudes4 <- ifelse(df2$Q15_6 == 1, "Agree", "Disagree")
df2$test_attitudes5 <- ifelse(df2$Q15_7 == 1, "Agree", "Disagree")
```

```
test_attitudes1_pval <- fisher.test(data.frame(
  rbind(table(df2[df2$ORA_REDEEMED == "Yes", ]$test_attitudes1),
    table(df2[df2$ORA_REDEEMED == "No", ]$test_attitudes1)),
  row.names = c("Ordered test kit", "Did not order test kit")
))
)p.value
```

```
test_attitudes2_pval <- fisher.test(data.frame(
  rbind(table(df2[df2$ORA_REDEEMED == "Yes", ]$test_attitudes2),
    table(df2[df2$ORA_REDEEMED == "No", ]$test_attitudes2)),
  row.names = c("Ordered test kit", "Did not order test kit")
))
)p.value
```

```
test_attitudes3_pval <- fisher.test(data.frame(
  rbind(table(df2[df2$ORA_REDEEMED == "Yes", ]$test_attitudes3),
    table(df2[df2$ORA_REDEEMED == "No", ]$test_attitudes3)),
  row.names = c("Ordered test kit", "Did not order test kit")
))
)p.value
```

```
test_attitudes4_pval <- fisher.test(data.frame(
  rbind(table(df2[df2$ORA_REDEEMED == "Yes", ]$test_attitudes4),
    table(df2[df2$ORA_REDEEMED == "No", ]$test_attitudes4)),
  row.names = c("Ordered test kit", "Did not order test kit")
))
)p.value
```

```
test_attitudes5_pval <- fisher.test(data.frame(
  rbind(table(df2[df2$ORA_REDEEMED == "Yes", ]$test_attitudes5),
    table(df2[df2$ORA_REDEEMED == "No", ]$test_attitudes5)),
  row.names = c("Ordered test kit", "Did not order test kit")
))
)p.value
```

```
attitudes <- list(
  "Getting tested for HIV helps people feel better" =
    list(var = "test_attitudes1", pval = test_attitudes1_pval),
```

```

"Getting tested for HIV helps people from getting HIV" =
  list(var = "test_attitudes2", pval = test_attitudes2_pval),
"People in my life would leave if I had HIV" =
  list(var = "test_attitudes3", pval = test_attitudes3_pval),
"People who tested positive for HIV should hide it from others" =
  list(var = "test_attitudes4", pval = test_attitudes4_pval),
"I would rather not know if I have HIV" =
  list(var = "test_attitudes5", pval = test_attitudes5_pval)
)

results_attitudes <- data.frame(
  Item = character(),
  Category = character(),
  Count_Yes = integer(),
  Percent_Yes = numeric(),
  Count_No = integer(),
  Percent_No = numeric(),
  P_value = character(),
  stringsAsFactors = FALSE
)

for (item in names(attitudes)) {
  var <- attitudes[[item]]$var
  p_val <- attitudes[[item]]$pval

  tab <- table(df2[[var]], df2$ORA_REDEEMED)
  first_row <- TRUE

  for (cat in c("Agree", "Disagree")) {
    count_yes <- ifelse(cat %in% rownames(tab), tab[cat, "Yes"], 0)
    count_no <- ifelse(cat %in% rownames(tab), tab[cat, "No"], 0)

    percent_yes <- round(100 * count_yes / sum(tab[, "Yes"]), 1)
    percent_no <- round(100 * count_no / sum(tab[, "No"]), 1)

    results_attitudes <- rbind(
      results_attitudes,
      data.frame(
        Item = if (first_row) item else "",
        Category = cat,
        Count_Yes = count_yes,
        Percent_Yes = percent_yes,
        Count_No = count_no,
        Percent_No = percent_no,
        P_value = if (first_row) round(p_val, 3) else "",
        stringsAsFactors = FALSE
      )
    )
    first_row <- FALSE
  }
}

kable(results_attitudes,
  col.names = c(

```

```

    "Attitude", "Response",
    "Ordered test kit", "Percent",
    "Did not order test kit", "Percent",
    "P-value"
  ))

```

| Attitude                                                      | Response | Ordered<br>test kit | Percent | Did not order<br>test kit | P-<br>Percent value |
|---------------------------------------------------------------|----------|---------------------|---------|---------------------------|---------------------|
| Getting tested for HIV helps people feel better               | Agree    | 169                 | 95.5    | 70                        | 90.9 0.16           |
|                                                               | Disagree | 8                   | 4.5     | 7                         | 9.1                 |
| Getting tested for HIV helps people from getting HIV          | Agree    | 148                 | 83.6    | 62                        | 81.6 0.717          |
|                                                               | Disagree | 29                  | 16.4    | 14                        | 18.4                |
| People in my life would leave if I had HIV                    | Agree    | 59                  | 33.7    | 37                        | 48.1 0.035          |
|                                                               | Disagree | 116                 | 66.3    | 40                        | 51.9                |
| People who tested positive for HIV should hide it from others | Agree    | 18                  | 10.2    | 9                         | 11.7 0.825          |
|                                                               | Disagree | 159                 | 89.8    | 68                        | 88.3                |
| I would rather not know if I have HIV                         | Agree    | 27                  | 15.3    | 15                        | 19.5 0.463          |
|                                                               | Disagree | 150                 | 84.7    | 62                        | 80.5                |

d) Attitudes toward human immunodeficiency virus (HIV) treatment (continuous scale from 1 [strongly disagree] to 7 [strongly agree])

```

hiv_trt_atts1_pval <- wilcox.test(
  Q94_1 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value

hiv_trt_atts2_pval <- wilcox.test(
  Q94_5 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value

hiv_trt_atts3_pval <- wilcox.test(
  Q94_6 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value

hiv_trt_atts4_pval <- wilcox.test(
  Q94_7 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value

hiv_trt_atts5_pval <- wilcox.test(

```

```

Q94_8 ~ ORA_REDEEMED,
data = df2,
exact = TRUE
)$p.value

hiv_trt_atts6_pval <- wilcox.test(
  Q94_9 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value

hiv_trt_atts7_pval <- wilcox.test(
  Q94_10 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value

hiv_trt_atts8_pval <- wilcox.test(
  Q94_11 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value

hiv_trt_atts9_pval <- wilcox.test(
  Q94_12 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value

hiv_trt_atts10_pval <- wilcox.test(
  Q94_13 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value

hiv_trt_items <- list(
  "I am less threatened by the idea of being HIV positive than I used to be" =
    list(var = "Q94_1", pval = hiv_trt_atts1_pval),
  "I am less worried about HIV infection than I used to be" =
    list(var = "Q94_5", pval = hiv_trt_atts2_pval),
  "I think HIV/AIDS is less of a problem than it used to be" =
    list(var = "Q94_6", pval = hiv_trt_atts3_pval),
  "I think HIV/AIDS is a less serious threat than it used to be because of new HIV/AIDS
  ↪ treatments" =
    list(var = "Q94_7", pval = hiv_trt_atts4_pval),
  "I am much less concerned about becoming HIV positive myself because of new HIV/AIDS
  ↪ treatments" =
    list(var = "Q94_8", pval = hiv_trt_atts5_pval),
  "I think that condom use during sex is less necessary now that new HIV/AIDS treatments
  ↪ are available" =
    list(var = "Q94_9", pval = hiv_trt_atts6_pval),
  "I think that someone who is HIV positive now needs to care less about condom use" =
    list(var = "Q94_10", pval = hiv_trt_atts7_pval),
  "I think that the need for condom use is less than it used to be, because you can
  ↪ always start new treatments" =

```

```

    list(var = "Q94_11", pval = hiv_trt_atts8_pval),
    "I think that someone who is HIV positive and uses new HIV/AIDS treatments can be
    ↪ cured" =
    list(var = "Q94_12", pval = hiv_trt_atts9_pval),
    "I think that new HIV/AIDS treatments can eradicate the virus from your body" =
    list(var = "Q94_13", pval = hiv_trt_atts10_pval)
)

# initialize results dataframe
results_hiv_trt <- data.frame(
  Statement = character(),
  Ordered_Test_Kit = character(),
  Did_Not_Order_Test_Kit = character(),
  P_value = character(),
  stringsAsFactors = FALSE
)

for (item in names(hiv_trt_items)) {
  var <- hiv_trt_items[[item]]$var
  p_val <- hiv_trt_items[[item]]$pval

  yes_vals <- df2[df2$ORA_REDEEMED == "Yes", var]
  no_vals <- df2[df2$ORA_REDEEMED == "No", var]

  yes_mean <- mean(yes_vals, na.rm = TRUE)
  yes_sd <- sd(yes_vals, na.rm = TRUE)

  no_mean <- mean(no_vals, na.rm = TRUE)
  no_sd <- sd(no_vals, na.rm = TRUE)

  results_hiv_trt <- rbind(
    results_hiv_trt,
    data.frame(
      Statement = item,
      Ordered_Test_Kit =
        paste0(round(yes_mean, 1), " (", round(yes_sd, 1), ")"),
      Did_Not_Order_Test_Kit =
        paste0(round(no_mean, 1), " (", round(no_sd, 1), ")"),
      P_value = round(p_val, 3),
      stringsAsFactors = FALSE
    )
  )
}

# display table
kable(
  results_hiv_trt,
  col.names = c(
    "Statements",
    "Ordered test kit: Mean (SD)",
    "Did not order test kit: Mean (SD)",
    "P-value"
  )
)

```

| Statements                                                                                                   | Ordered test<br>kit: Mean (SD) | Did not order test<br>kit: Mean (SD) | P-<br>value |
|--------------------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------------|-------------|
| I am less threatened by the idea of being HIV positive than I used to be                                     | 4.1 (2)                        | 3.8 (2.2)                            | 0.421       |
| I am less worried about HIV infection than I used to be                                                      | 3.7 (2)                        | 3.5 (1.8)                            | 0.400       |
| I think HIV/AIDS is less of a problem than it used to be                                                     | 3.9 (2.1)                      | 3.6 (2.1)                            | 0.413       |
| I think HIV/AIDS is a less serious threat than it used to be because of new HIV/AIDS treatments              | 4.2 (2)                        | 3.8 (2.1)                            | 0.225       |
| I am much less concerned about becoming HIV positive myself because of new HIV/AIDS treatments               | 3.4 (2)                        | 3 (1.9)                              | 0.274       |
| I think that condom use during sex is less necessary now that new HIV/AIDS treatments are available          | 2.4 (1.8)                      | 2.2 (1.5)                            | 0.971       |
| I think that someone who is HIV positive now needs to care less about condom use                             | 1.9 (1.7)                      | 2.5 (2.1)                            | 0.064       |
| I think that the need for condom use is less than it used to be, because you can always start new treatments | 2.5 (1.9)                      | 2.5 (2)                              | 0.767       |
| I think that someone who is HIV positive and uses new HIV/AIDS treatments can be cured                       | 3.2 (1.9)                      | 3.6 (2.2)                            | 0.199       |
| I think that new HIV/AIDS treatments can eradicate the virus from your body                                  | 3.1 (1.9)                      | 3.8 (2.1)                            | 0.029       |

#### e) Human immunodeficiency virus (HIV)-related stigma among study participants

```
hiv_stigma1_pval <- wilcox.test(
  Q14_2 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value
```

```
hiv_stigma2_pval <- wilcox.test(
  Q14_3 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value
```

```
hiv_stigma3_pval <- wilcox.test(
  Q14_4 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value
```

```
hiv_stigma4_pval <- wilcox.test(
  Q14_5 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value
```

```
likert_levels <- c(
  "Strongly agree", "Agree", "Somewhat agree",
  "Neither agree nor disagree",
```

```

    "Somewhat disagree", "Disagree", "Strongly disagree"
  )

  stigma_items <- list(
    "I feel afraid of people living with HIV/AIDS" =
      list(var = "Q14_2", pval = hiv_stigma1_pval),
    "I could not be friends with someone who has HIV/AIDS" =
      list(var = "Q14_3", pval = hiv_stigma2_pval),
    "People who get HIV/AIDS through sex or drug use got what they deserve" =
      list(var = "Q14_4", pval = hiv_stigma3_pval),
    "I feel anger toward people with HIV/AIDS" =
      list(var = "Q14_5", pval = hiv_stigma4_pval)
  )

  results_stigma <- data.frame(
    Item = character(),
    Category = character(),
    Count_Yes = integer(),
    Percent_Yes = numeric(),
    Count_No = integer(),
    Percent_No = numeric(),
    P_value = character(),
    stringsAsFactors = FALSE
  )

  for (item in names(stigma_items)) {
    var <- stigma_items[[item]]$var
    p_val <- stigma_items[[item]]$pval

    tab <- table(factor(df2[[var]], levels = 1:7, labels = likert_levels),
                  df2$ORA_REDEEMED)

    first_row <- TRUE
    for (cat in likert_levels) {
      count_yes <- ifelse(cat %in% rownames(tab), tab[cat, "Yes"], 0)
      count_no <- ifelse(cat %in% rownames(tab), tab[cat, "No"], 0)

      percent_yes <- round(100 * count_yes / sum(tab[, "Yes"]), 1)
      percent_no <- round(100 * count_no / sum(tab[, "No"]), 1)

      results_stigma <- rbind(
        results_stigma,
        data.frame(
          Item = if (first_row) item else "",
          Category = cat,
          Count_Yes = count_yes,
          Percent_Yes = percent_yes,
          Count_No = count_no,
          Percent_No = percent_no,
          P_value = if (first_row) round(p_val, 3) else "",
          stringsAsFactors = FALSE
        )
      )
    }
    first_row <- FALSE
  }

```

```

    }
  }

kable(results_stigma,
      col.names = c(
        "HIV stigma", "Response",
        "Ordered test kit", "Percent",
        "Did not order test kit", "Percent",
        "P-value"
      ))

```

| HIV stigma                                                               | Response                   | Ordered<br>test kit | Percent | Did not<br>order test kit | Percent | P-<br>value |
|--------------------------------------------------------------------------|----------------------------|---------------------|---------|---------------------------|---------|-------------|
| I feel afraid of people living with<br>HIV/AIDS                          | Strongly agree             | 12                  | 6.8     | 7                         | 9.1     | 0.613       |
|                                                                          | Agree                      | 9                   | 5.1     | 2                         | 2.6     |             |
|                                                                          | Somewhat agree             | 21                  | 11.9    | 9                         | 11.7    |             |
|                                                                          | Neither agree nor disagree | 20                  | 11.3    | 9                         | 11.7    |             |
|                                                                          | Somewhat disagree          | 11                  | 6.2     | 6                         | 7.8     |             |
|                                                                          | Disagree                   | 32                  | 18.1    | 17                        | 22.1    |             |
|                                                                          | Strongly disagree          | 72                  | 40.7    | 27                        | 35.1    |             |
|                                                                          |                            |                     |         |                           |         |             |
| I could not be friends with someone<br>who has HIV/AIDS                  | Strongly agree             | 1                   | 0.6     | 4                         | 5.2     | 0.032       |
|                                                                          | Agree                      | 1                   | 0.6     | 0                         | 0.0     |             |
|                                                                          | Somewhat agree             | 3                   | 1.7     | 1                         | 1.3     |             |
|                                                                          | Neither agree nor disagree | 9                   | 5.1     | 6                         | 7.8     |             |
|                                                                          | Somewhat disagree          | 7                   | 4.0     | 1                         | 1.3     |             |
|                                                                          | Disagree                   | 26                  | 14.7    | 19                        | 24.7    |             |
|                                                                          | Strongly disagree          | 130                 | 73.4    | 46                        | 59.7    |             |
|                                                                          |                            |                     |         |                           |         |             |
| People who get HIV/AIDS through sex<br>or drug use got what they deserve | Strongly agree             | 2                   | 1.1     | 0                         | 0.0     | 0.332       |
|                                                                          | Agree                      | 2                   | 1.1     | 1                         | 1.3     |             |
|                                                                          | Somewhat agree             | 3                   | 1.7     | 2                         | 2.6     |             |
|                                                                          | Neither agree nor disagree | 8                   | 4.5     | 7                         | 9.1     |             |
|                                                                          | Somewhat disagree          | 6                   | 3.4     | 2                         | 2.6     |             |
|                                                                          | Disagree                   | 24                  | 13.6    | 12                        | 15.6    |             |
|                                                                          | Strongly disagree          | 132                 | 74.6    | 53                        | 68.8    |             |
|                                                                          |                            |                     |         |                           |         |             |
| I feel anger toward people with<br>HIV/AIDS                              | Strongly agree             | 1                   | 0.6     | 0                         | 0.0     | 0.213       |
|                                                                          | Agree                      | 0                   | 0.0     | 0                         | 0.0     |             |

| HIV stigma | Response                   | Ordered<br>test kit | Percentorder | Did not<br>test kit | P-<br>Percentvalue |
|------------|----------------------------|---------------------|--------------|---------------------|--------------------|
|            | Somewhat agree             | 0                   | 0.0          | 0                   | 0.0                |
|            | Neither agree nor disagree | 11                  | 6.2          | 5                   | 6.5                |
|            | Somewhat disagree          | 3                   | 1.7          | 0                   | 0.0                |
|            | Disagree                   | 19                  | 10.7         | 16                  | 20.8               |
|            | Strongly disagree          | 143                 | 80.8         | 56                  | 72.7               |

## f) Medical mistrust

```

med_mistrust1_pval <- wilcox.test(
  Q16_1 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value

med_mistrust2_pval <- wilcox.test(
  Q16_2 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value

med_mistrust3_pval <- wilcox.test(
  Q16_3 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value

med_mistrust4_pval <- wilcox.test(
  Q16_4 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value

med_mistrust5_pval <- wilcox.test(
  Q16_5 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value

med_mistrust6_pval <- wilcox.test(
  Q16_6 ~ ORA_REDEEMED,
  data = df2,
  exact = TRUE
)$p.value

med_mistrust7_pval <- wilcox.test(
  Q16_7 ~ ORA_REDEEMED,

```

```

data = df2,
exact = TRUE
)$p.value

mistrust_levels <- c(
  "Strongly agree", "Agree", "Disagree", "Strongly disagree"
)

mistrust_items <- list(
  "You'd better be cautious when dealing with health care organizations" =
    list(var = "Q16_1", pval = med_mistrust1_pval),
  "Patients have sometimes been deceived or misled by health care organizations" =
    list(var = "Q16_2", pval = med_mistrust2_pval),
  "When health care organizations make mistakes they usually cover it up" =
    list(var = "Q16_3", pval = med_mistrust3_pval),
  "Health care organizations have sometimes done harmful experiments on patients without
    ↪ their knowledge" =
    list(var = "Q16_4", pval = med_mistrust4_pval),
  "Health care organizations don't always keep your information totally private" =
    list(var = "Q16_5", pval = med_mistrust5_pval),
  "Sometimes I wonder if health care organizations really know what they are doing" =
    list(var = "Q16_6", pval = med_mistrust6_pval),
  "Mistakes are common in health care organizations" =
    list(var = "Q16_7", pval = med_mistrust7_pval)
)

results_mistrust <- data.frame(
  Item = character(),
  Category = character(),
  Count_Yes = integer(),
  Percent_Yes = numeric(),
  Count_No = integer(),
  Percent_No = numeric(),
  P_value = character(),
  stringsAsFactors = FALSE
)

for (item in names(mistrust_items)) {
  var <- mistrust_items[[item]]$var
  p_val <- mistrust_items[[item]]$pval

  if(item ==
    "You'd better be cautious when dealing with health care organizations") {
    level = c(28, 30, 33, 34)
  }
  else {
    level = c(1, 2, 6, 7)
  }

  tab <- table(
    factor(
      df2[[var]],
      levels = level,

```

```

      labels = c(
        "Strongly agree",
        "Agree",
        "Disagree",
        "Strongly disagree"
      )
    ),
    df2$ORA_REDEEMED,
    useNA = "no"
  )

first_row <- TRUE
for (cat in mistrust_levels) {
  count_yes <- ifelse(cat %in% rownames(tab), tab[cat, "Yes"], 0)
  count_no  <- ifelse(cat %in% rownames(tab), tab[cat, "No"], 0)

  percent_yes <- round(100 * count_yes / sum(tab[, "Yes"]), 1)
  percent_no  <- round(100 * count_no / sum(tab[, "No"]), 1)

  results_mistrust <- rbind(
    results_mistrust,
    data.frame(
      Item = if (first_row) item else "",
      Category = cat,
      Count_Yes = count_yes,
      Percent_Yes = percent_yes,
      Count_No = count_no,
      Percent_No = percent_no,
      P_value = if (first_row) round(p_val, 3) else "",
      stringsAsFactors = FALSE
    )
  )
  first_row <- FALSE
}
}

kable(results_mistrust,
      col.names = c(
        "Medical mistrust", "Response",
        "Ordered test kit", "Percent",
        "Did not order test kit", "Percent",
        "P-value"
      ))

```

| Medical mistrust                                                        | Response          | Ordered<br>test kit | Percent | Did not<br>order test<br>kit | P-<br>Percent | value |
|-------------------------------------------------------------------------|-------------------|---------------------|---------|------------------------------|---------------|-------|
| You'd better be cautious when dealing with<br>health care organizations | Strongly agree    | 37                  | 20.9    | 17                           | 22.4          | 0.503 |
|                                                                         | Agree             | 66                  | 37.3    | 32                           | 42.1          |       |
|                                                                         | Disagree          | 42                  | 23.7    | 14                           | 18.4          |       |
|                                                                         | Strongly disagree | 32                  | 18.1    | 13                           | 17.1          |       |

| Medical mistrust                                                                                      | Response          | Ordered<br>test kit | Percent | Did not<br>order test<br>kit | P-<br>Percent | value |
|-------------------------------------------------------------------------------------------------------|-------------------|---------------------|---------|------------------------------|---------------|-------|
| Patients have sometimes been deceived or misled by health care organizations                          | Strongly agree    | 32                  | 18.2    | 11                           | 14.3          | 0.413 |
|                                                                                                       | Agree             | 83                  | 47.2    | 34                           | 44.2          |       |
|                                                                                                       | Disagree          | 33                  | 18.8    | 22                           | 28.6          |       |
|                                                                                                       | Strongly disagree | 28                  | 15.9    | 10                           | 13.0          |       |
| When health care organizations make mistakes they usually cover it up                                 | Strongly agree    | 29                  | 16.7    | 10                           | 13.2          | 0.222 |
|                                                                                                       | Agree             | 83                  | 47.7    | 29                           | 38.2          |       |
|                                                                                                       | Disagree          | 36                  | 20.7    | 30                           | 39.5          |       |
|                                                                                                       | Strongly disagree | 26                  | 14.9    | 7                            | 9.2           |       |
| Health care organizations have sometimes done harmful experiments on patients without their knowledge | Strongly agree    | 30                  | 17.0    | 14                           | 18.2          | 0.413 |
|                                                                                                       | Agree             | 73                  | 41.5    | 25                           | 32.5          |       |
|                                                                                                       | Disagree          | 51                  | 29.0    | 26                           | 33.8          |       |
|                                                                                                       | Strongly disagree | 22                  | 12.5    | 12                           | 15.6          |       |
| Health care organizations don't always keep your information totally private                          | Strongly agree    | 36                  | 20.6    | 11                           | 14.5          | 0.371 |
|                                                                                                       | Agree             | 64                  | 36.6    | 29                           | 38.2          |       |
|                                                                                                       | Disagree          | 47                  | 26.9    | 23                           | 30.3          |       |
|                                                                                                       | Strongly disagree | 28                  | 16.0    | 13                           | 17.1          |       |
| Sometimes I wonder if health care organizations really know what they are doing                       | Strongly agree    | 22                  | 12.5    | 9                            | 11.7          | 0.965 |
|                                                                                                       | Agree             | 66                  | 37.5    | 28                           | 36.4          |       |
|                                                                                                       | Disagree          | 60                  | 34.1    | 30                           | 39.0          |       |
|                                                                                                       | Strongly disagree | 28                  | 15.9    | 10                           | 13.0          |       |
| Mistakes are common in health care organizations                                                      | Strongly agree    | 21                  | 11.9    | 8                            | 10.7          | 0.638 |
|                                                                                                       | Agree             | 89                  | 50.6    | 36                           | 48.0          |       |
|                                                                                                       | Disagree          | 46                  | 26.1    | 23                           | 30.7          |       |
|                                                                                                       | Strongly disagree | 20                  | 11.4    | 8                            | 10.7          |       |

**Figure S1**

```
library(ggplot2)
```

```
q7_vars <- c(
  "Q7_10_FU1", "Q7_11_FU1", "Q7_12_FU1", "Q7_13_FU1",
  "Q7_14_FU1", "Q7_15_FU1", "Q7_16_FU1", "Q7_17_FU1",
  "Q7_18_FU1", "Q7_19_FU1", "Q7_2_FU1", "Q7_3_FU1",
  "Q7_4_FU1", "Q7_5_FU1", "Q7_6_FU1", "Q7_7_FU1",
```

```

"Q7_8_FU1", "Q7_9_FU1"
)

question_labels <- c(
  Q7_10_FU1 = "I am ashamed to tell others that I am on PrEP.",
  Q7_11_FU1 = "Not knowing if there are long-term side effects of taking a daily HIV pill
  ↪ makes me very uncomfortable.",
  Q7_12_FU1 = "I would be more comfortable using PrEP if I just knew how it would affect
  ↪ my health.",
  Q7_13_FU1 = "PrEP is too expensive.",
  Q7_14_FU1 = "Gay and bisexual men who take PrEP are being responsible.",
  Q7_15_FU1 = "PrEP is an excuse for gay and bisexual men to avoid using condoms.",
  Q7_16_FU1 = "Gay and bisexual men who take PrEP are risk-takers.",
  Q7_17_FU1 = "PrEP gives gay and bisexual men more options to remain safe.",
  Q7_18_FU1 = "Gay and bisexual men who take PrEP are promiscuous.",
  Q7_19_FU1 = "PrEP is only partially effective.",
  Q7_2_FU1 = "Taking a daily HIV pill is a good way to protect myself from getting
  ↪ HIV.",
  Q7_3_FU1 = "By taking PrEP, I am lowering my chances of getting infected with HIV.",
  Q7_4_FU1 = "Taking PrEP means I can have sex without using condoms.",
  Q7_5_FU1 = "Having PrEP available makes safer sex less important.",
  Q7_6_FU1 = "Taking PrEP makes me more likely to have sex without using a condom.",
  Q7_7_FU1 = "I feel uncomfortable taking HIV medication when I don't have HIV.",
  Q7_8_FU1 = "I worry that others will think I have HIV if they find out I am on PrEP.",
  Q7_9_FU1 = "I worry about what other people would think of me if they knew I was on
  ↪ PrEP."
)

response_labels <- c(
  "1" = "Strongly agree",
  "2" = "Somewhat agree",
  "3" = "Neither agree or disagree",
  "4" = "Somewhat disagree",
  "5" = "Strongly disagree"
)

df_s1 <- df %>%
  select(all_of(q7_vars)) %>%
  pivot_longer(
    cols = everything(),
    names_to = "Question",
    values_to = "Response"
  ) %>%
  filter(Response != 9999) %>%
  mutate(
    Response = factor(
      Response,
      levels = 1:5,
      labels = response_labels
    )
  ) %>%
  count(Question, Response) %>%
  group_by(Question) %>%
  mutate(

```

```

percent = n / sum(n),
label = scales::percent(percent, accuracy = 0.01)
) %>%
ungroup()

```

```

ggplot(df_s1, aes(y = Question, x = percent, fill = Response)) +
  geom_col() +
  geom_text(
    aes(label = label),
    position = position_stack(vjust = 0.5),
    size = 5,
    color = "black"
  ) +
  scale_x_continuous(labels = scales::percent) +
  scale_y_discrete(labels = question_labels) +
  scale_fill_brewer(palette = "RdYlBu", direction = -1) +
  labs(
    y = "",
    x = "",
    fill = "Response",
    title="Figure S1 from Appendix 3"
  ) +
  theme(
    plot.title = element_text(size = 24),
    axis.text.y = element_text(size = 20),
    panel.grid.major.y = element_blank(),
    legend.text = element_text(size = 20),
    legend.title = element_text(size = 20)
  )

```

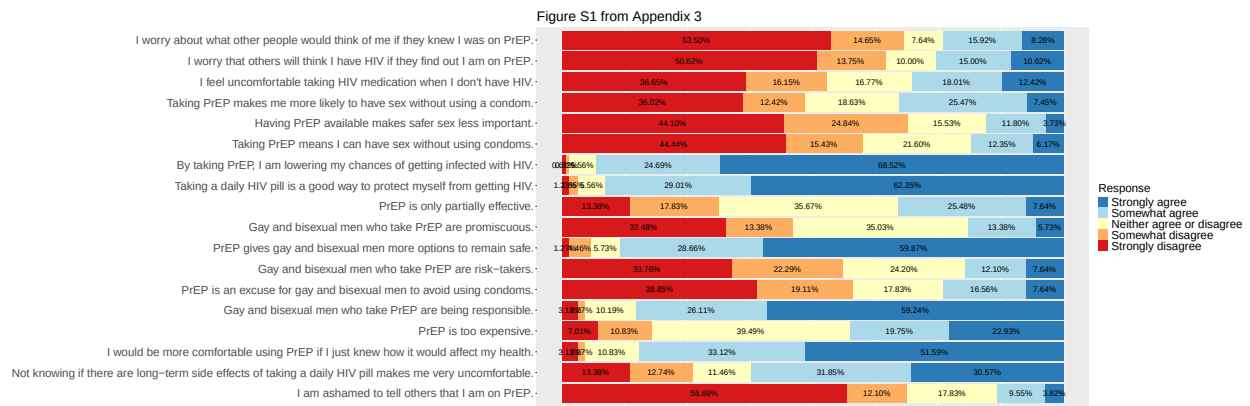


Figure S2

```

q8_1_vars <- c(
  "Q8_1_10_FU1", "Q8_1_11_FU1", "Q8_1_12_FU1", "Q8_1_13_FU1",
  "Q8_1_1_FU1", "Q8_1_2_FU1", "Q8_1_3_FU1", "Q8_1_4_FU1",
  "Q8_1_5_FU1", "Q8_1_6_FU1", "Q8_1_7_FU1", "Q8_1_8_FU1",
  "Q8_1_9_FU1"
)

```

```

)

q8_1_labels <- c(
  Q8_1_10_FU1 = "Potentially having long-term health effects because of PrEP",
  Q8_1_11_FU1 = "If I do become HIV+, certain medicines won't work because I was taking
    ↪ PrEP",
  Q8_1_12_FU1 = "I do not really think I am at high enough risk for HIV to warrant taking
    ↪ PrEP",
  Q8_1_13_FU1 = "I would be ashamed to take PrEP",
  Q8_1_1_FU1  = "Having to take a pill every day",
  Q8_1_2_FU1  = "Potentially experiencing side effects",
  Q8_1_3_FU1  = "PrEP not protecting me against HIV infection completely",
  Q8_1_4_FU1  = "PrEP might make me more willing to have anal sex without a condom",
  Q8_1_5_FU1  = "Taking PrEP might make my partner(s) expect me to have anal sex without
    ↪ a condom",
  Q8_1_6_FU1  = "Having to take PrEP means I'm putting myself at risk for HIV",
  Q8_1_7_FU1  = "People will see me taking the medication and will want to know why I'm
    ↪ taking it",
  Q8_1_8_FU1  = "People will see me taking the medication and think that I have HIV",
  Q8_1_9_FU1  = "I don't want to talk to my doctor about my sex life"
)

q8_1_response_labels <- c(
  "1" = "Not at all important",
  "2" = "Slightly important",
  "3" = "Moderately important",
  "4" = "Very important",
  "5" = "Extremely important"
)

df_q8_1 <- df %>%
  select(all_of(q8_1_vars)) %>%
  pivot_longer(
    cols = everything(),
    names_to = "Question",
    values_to = "Response"
  ) %>%
  filter(Response != 9999) %>%
  mutate(
    Response = factor(
      Response,
      levels = 1:5,
      labels = q8_1_response_labels
    )
  ) %>%
  count(Question, Response) %>%
  group_by(Question) %>%
  mutate(
    percent = n / sum(n),
    label = scales::percent(percent, accuracy = 0.01)
  ) %>%
  ungroup()

```

```

ggplot(df_q8_1, aes(y = Question, x = percent, fill = Response)) +
  geom_col() +
  geom_text(
    aes(label = label),
    position = position_stack(vjust = 0.5),
    size = 5,
    color = "black"
  ) +
  scale_x_continuous(labels = scales::percent) +
  scale_y_discrete(labels = q8_1_labels) +
  scale_fill_brewer(palette = "RdYlBu", direction = -1) +
  labs(
    y = NULL,
    x = "Percent",
    fill = "Importance",
    title="Figure S2 from Appendix 3"
  ) +
  theme(
    plot.title = element_text(size = 24),
    axis.text.y = element_text(size = 20),
    panel.grid.major.y = element_blank(),
    legend.text = element_text(size = 20),
    legend.title = element_text(size = 20)
  )

```

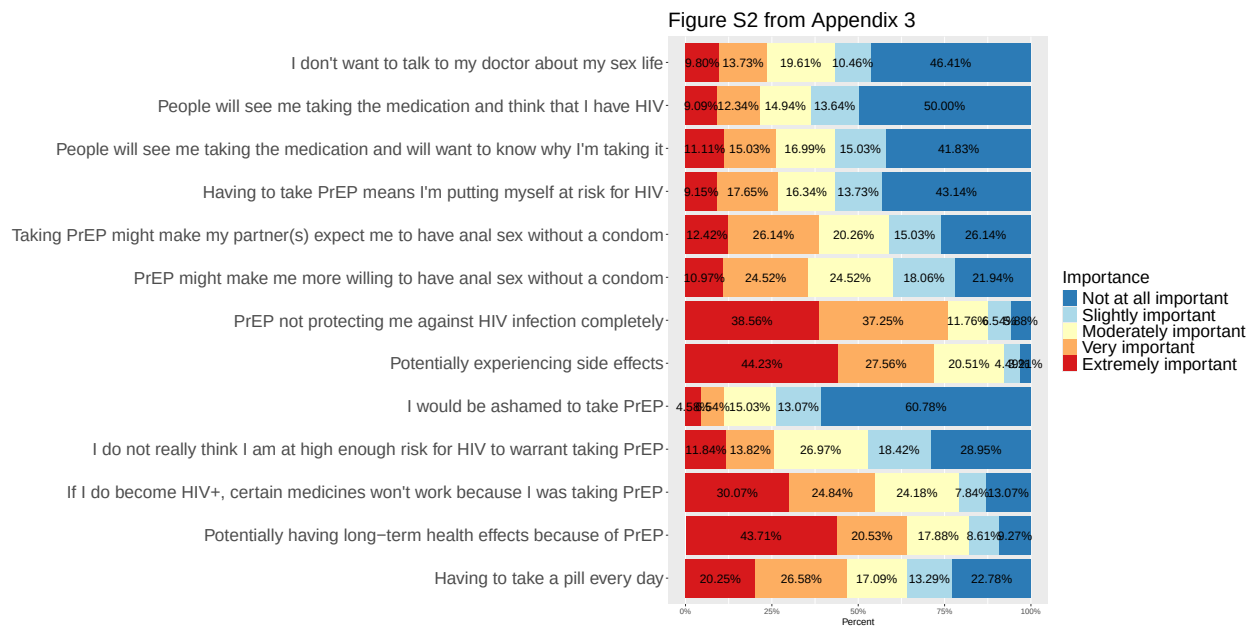


Figure S3

```

q8_2_vars <- c(
  "Q8_2_1_FU1", "Q8_2_2_FU1", "Q8_2_3_FU1", "Q8_2_4_FU1",
  "Q8_2_5_FU1", "Q8_2_6_FU1", "Q8_2_7_FU1", "Q8_2_8_FU1"
)

```

```

q8_2_labels <- c(
  Q8_2_1_FU1 = "Not having to pay for PrEP or because it is affordable",
  Q8_2_2_FU1 = "Getting free HIV/STD testing while I am on PrEP",
  Q8_2_3_FU1 = "Getting free one-on-one counseling and support while I am on PrEP",
  Q8_2_4_FU1 = "Not having to go to my regular doctor to get PrEP",
  Q8_2_5_FU1 = "Hearing from my doctor that taking PrEP would be best for my health",
  Q8_2_6_FU1 = "I am concerned that, even if I use condoms, I could still get HIV",
  Q8_2_7_FU1 = "I am a gay/bisexual guy or a transwoman and I know I am at really high
  ↪ risk for HIV",
  Q8_2_8_FU1 = "Hearing from my friends that they had good experiences with PrEP"
)

```

```

q8_2_response_labels <- c(
  "1" = "Extremely important",
  "2" = "Very important",
  "3" = "Moderately important",
  "4" = "Slightly important",
  "5" = "Not at all important"
)

```

```

df_q8_2 <- df %>%
  select(all_of(q8_2_vars)) %>%
  pivot_longer(
    cols = everything(),
    names_to = "Question",
    values_to = "Response"
  ) %>%
  filter(Response != 9999) %>%
  mutate(
    Response = factor(
      Response,
      levels = 1:5,
      labels = q8_2_response_labels
    )
  ) %>%
  count(Question, Response) %>%
  group_by(Question) %>%
  mutate(
    percent = n / sum(n),
    label = scales::percent(percent, accuracy = 0.01)
  ) %>%
  ungroup()

```

```

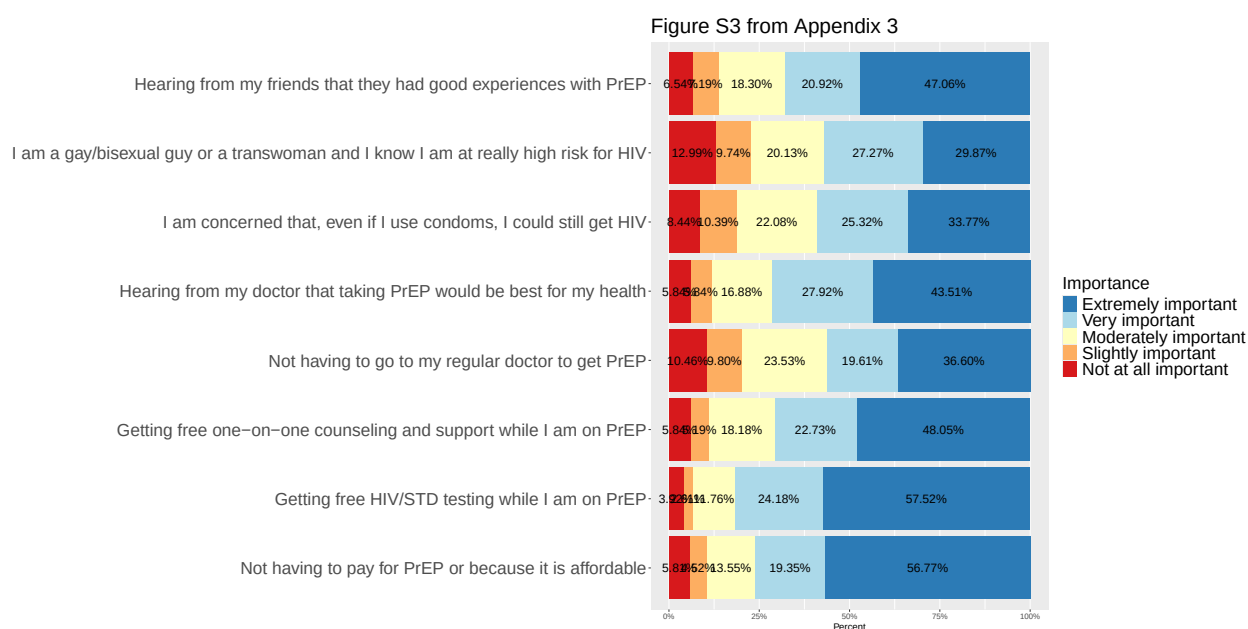
ggplot(df_q8_2, aes(y = Question, x = percent, fill = Response)) +
  geom_col() +
  geom_text(
    aes(label = label),
    position = position_stack(vjust = 0.5),
    size = 5,
    color = "black"
  ) +
  scale_x_continuous(labels = scales::percent) +
  scale_y_discrete(labels = q8_2_labels) +

```

```

scale_fill_brewer(palette = "RdYlBu", direction = -1) +
labs(
  y = NULL,
  x = "Percent",
  fill = "Importance",
  title="Figure S3 from Appendix 3"
) +
theme(
  plot.title = element_text(size = 24),
  axis.text.y = element_text(size = 20),
  panel.grid.major.y = element_blank(),
  legend.text = element_text(size = 20),
  legend.title = element_text(size = 20)
)

```



## Reflection

### Primary Analysis

Table 1 can be almost perfectly replicated, with a few small differences and notes:

- The race section's counts and percentages are out of 250 instead of 254, as there are 4 participants whose race was left as NA so they were excluded.
- There was one individual who took the HIV test but also marked a reason not to take it, so they were excluded from the "reasons not to take HIV test" section.
- The IQR for the age in years, as well as the median and IQR for months since last HIV test are slightly different from the original Table 1.

As for the contrast significance testing, the p-values obtained are slightly different from the original, although the conclusions on significance are the same, with all the platforms in Wave 1 not being significantly different

from each other at the 0.05 level, and all the platforms in Wave 2 being significantly different from each other (mainly due to Jack'd having a lot more test kits ordered than any other platform). However, due to Bing having exactly zero test kits ordered, with the original model, the p-values for both of Wave 2's contrasts that involve Bing are highly non-significant. This may be due to slight differences in how R and SAS calculate the test statistics. However, by adding a small pseudocount of 0.5 to Bing's test kit number to stabilize the estimate, the p-values flip from being insignificant to a similar significance as the paper, with the contrasts for Bing vs Instagram being slightly less significant than Bing vs Jack'd or Jack'd vs Instagram.

## Secondary Analysis

The secondary analysis, depicted in Appendix 3, is also able to be almost perfectly replicated. The only difference is that for the first table on alcohol/substance use, I was unable to find and account for the 5 rows for the sedative and prescribed stimulants category that were listed as missing, so my counts and p-values were slightly off.

## References

Manuscript of the original paper and its appendices, which were followed to reproduce the results, can be found here: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9591705/>

<https://bookdown.org/yihui/rmarkdown-cookbook/text-width.html>

<https://www.rdocumentation.org/packages/stats/versions/3.6.2/topics/wilcox.test>

ChatGPT was used to generate some sections of this code as well as for some editing and debugging.